

ChronoSolve

Generate Once. Adapt Intelligently. Disrupt Minimally.

Problem Statement	PS1 - Smart Timetable Generator
Project	ChronoSolve - Adaptive Constraint-Optimized Academic Scheduling
Team Name	4L's
Use Case Scope	General higher-education academic scheduling (college / university departments)
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1. Problem Statement & Context

PS1 - Smart Timetable Generator. Across colleges and universities, timetable coordinators must reconcile faculty availability, rooms and laboratories, batches, electives, session durations, contact hours, lunch periods, maximum consecutive teaching hours, and workload rules. These constraints are tightly coupled: changing one assignment can create conflicts elsewhere, making spreadsheet-driven generation and revision difficult to manage. ChronoSolve is intentionally institution-agnostic, so the same engine can accept configurable academic rules instead of depending on one college or department.

The scale is combinatorial: an illustrative department with 150 weekly sessions, 40 timeslots, and 20 rooms can expose up to 120,000 session-slot-room candidate assignments before constraints are applied, while the number of complete assignment combinations is vastly larger. Lab blocks, parallel electives, shared faculty, and room restrictions couple these choices, making exhaustive enumeration impractical.

The challenge becomes more critical after publication: faculty leave, room maintenance, events, or changed academic requirements can invalidate only part of an otherwise usable timetable. Regenerating everything causes unnecessary disruption, while manual patching risks new clashes. ChronoSolve addresses both initial conflict-free generation and stable repair after real-world changes.

2. Proposed Solution & Constraint Model

ChronoSolve is an adaptive academic scheduling platform built around Google OR-Tools CP-SAT. For the prototype, each feasible (session, timeslot, room) candidate is represented by a Boolean decision variable, with exactly one valid placement selected per session. Clash-freedom is expressed through $\text{sum} \leq 1$ constraints for each faculty, room, and batch at a timeslot; capacity, room/lab type, and unavailability remove forbidden assignments; multi-hour labs are restricted to valid contiguous blocks that do not cross lunch or day boundaries. Additional institutional rules are encoded as hard constraints, while preferences become weighted soft penalties.

A coordinator imports department data from Excel, validates it, configures rules, and generates synchronized batch, faculty, and room timetables. When a disruption occurs, the previous published assignments enter the optimization objective: CP-SAT restores feasibility while penalizing unnecessary changes, producing a revised timetable that preserves as much of the existing schedule as possible.

3. Target Users

Primary users are timetable coordinators and Heads of Department. Faculty provide availability/preferences and access personal schedules; students access the current batch timetable and revisions; academic administrators review faculty load, room utilization, feasibility, and scheduling performance.

4. Key Features

- **Constraint-Guaranteed Generation:** Prevents faculty, room, and batch double-booking; enforces room capacity/type, faculty availability, weekly contact hours, lunch protection, contiguous labs, maximum consecutive teaching hours, daily/weekly load limits, and parallel elective requirements.
- **Tunable Multi-Objective Optimization:** Balances student gaps, faculty idle windows, daily load, subject spread, and institutional preferences using weighted soft objectives whose weights the coordinator can adjust and re-solve against.

- **Minimum-Disruption Re-Optimization - Core Innovation:** Repairs faculty/room disruptions while minimizing deviations from the published timetable. Changed sessions are highlighted and schedule retention = unchanged sessions / total sessions is reported.
- **What-If Simulation:** Simulates faculty leave, room/lab unavailability, or rule changes and previews affected sessions and proposed repairs before approval.
- **Natural-Language Constraint Entry:** Common instructions such as 'Prof. Mehta is unavailable after 2 PM on Friday' are converted into structured candidate rules, checked against known entities, previewed, and confirmed before solver execution.
- **Constraint-Aware Explainability:** For a selected session or candidate alternative, ChronoSolve evaluates the active hard constraints and reports the specific faculty, batch, room, capacity, availability, or institutional rule blocking that placement. Explanations use the same structured constraint data that builds the solver model.
- **Infeasibility Diagnosis:** If no feasible schedule exists, targeted constraint checks identify relevant competing rule categories and present actionable relaxation suggestions instead of returning an invalid timetable.
- **Synchronized Views & Outputs:** Provides batch/faculty/room views, revision highlighting, solve-progress feedback, analytics, and Excel/PDF/calendar export.

5. Innovation & Differentiation

Conventional rule-checking or heuristic scheduling approaches can generate or validate a timetable, but post-publication changes create a different requirement: the coordinator needs the smallest practical valid repair, not an unrelated fresh schedule. ChronoSolve makes schedule stability part of the optimization objective itself. The published timetable is used as the reference, deviations are penalized, and the repaired schedule reports retention numerically. What-if analysis, constraint-aware reasoning, and controlled natural-language rule entry make that optimization workflow understandable and usable by non-technical coordinators.

6. Technology Stack, Cost & Scalability

- **Frontend:** React + Tailwind CSS.
- **Backend & Data:** Python, FastAPI, and PostgreSQL.
- **Optimization:** Google OR-Tools CP-SAT.
- **Data / Export:** pandas, openpyxl, ReportLab, and ICS.
- **Deployment & Cost:** ₹0 hackathon prototype using open-source/free tools, local PostgreSQL and solver execution, synthetic or authorized anonymized data, and available free-tier hosting for an online demo. The complete project remains locally executable as a presentation fallback and has no mandatory paid dependency.

The web, data, and solver layers are modular. Larger deployments can partition unrelated academic units by department/year/constraint groups, while disruption repair can focus on affected sessions and connected dependencies. Scalability will be evaluated through measured solve time, feasibility, and objective quality on progressively larger test datasets.

7. Expected Impact & Evaluation

ChronoSolve aims to reduce repeated manual conflict reconciliation, improve timetable feasibility, and make post-publication changes safer. Prototype evaluation will report generation/repair time, hard-constraint violations, student idle time, faculty idle windows/load balance, room utilization, solver time/objective quality, and schedule retention after disruptions. Reported figures will come from the implemented prototype rather than projections.

Because the core scheduling engine requires no paid proprietary service, the hackathon prototype demonstrates a practical zero-cost adoption path while remaining extensible for larger institutional deployment.

8. Working Prototype Demonstration

End-to-end finale workflow

01 IMPORT & VALIDATE Excel data, faculty, rooms, labs & rules	02 GENERATE CP-SAT creates a feasible timetable	03 INSPECT VIEWS Batch, faculty & room schedules	04 SIMULATE DISRUPTION Faculty / room / lab becomes unavailable
05 WHAT-IF ANALYSIS Preview affected sessions before applying	06 RE-OPTIMIZE Repair while preserving published assignments	07 EXPLAIN & MEASURE Changed cells, retention % & constraint reasons	08 APPROVE & EXPORT Publish / export Excel, PDF or calendar

ChronoSolve: conflict-free generation when planning begins, minimum-disruption repair when reality changes.